Fackverk och linjära ekvationssystem

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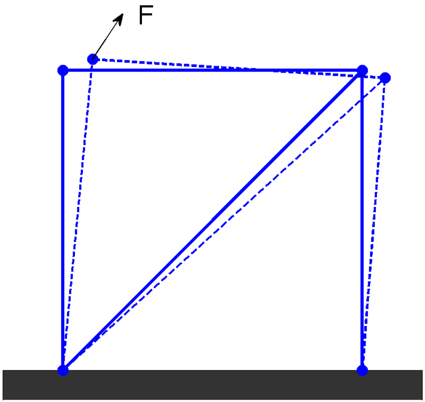
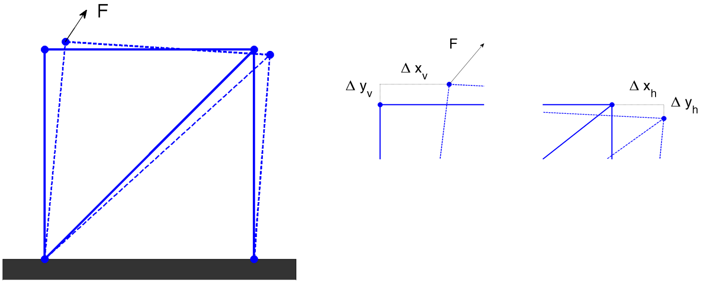
Fackverk



- Fackverk: 2 fixa noder, 2 rörliga noder, 4 stänger.
- Kraften **F** appliceras i vänstra noden
- $\Rightarrow$ Deformation.

Fackverk

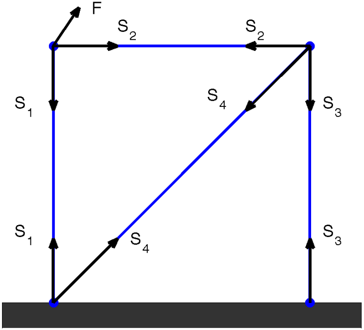


- Givet $\mathbf{F} = (F_x, F_y)$, beräkna $(\Delta x_v, \Delta y_v)$ och $(\Delta x_h, \Delta y_h)$.
- $\Rightarrow$ Linjärt ekvationssystem

$$\begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 1 + \frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} \\ 0 & 0 & \frac{1}{2\sqrt{2}} & 1 + \frac{1}{2\sqrt{2}} \end{pmatrix} \begin{pmatrix} \Delta x_v \\ \Delta y_v \\ \Delta x_h \\ \Delta y_h \end{pmatrix} = \begin{pmatrix} F_x \\ F_y \\ 0 \\ 0 \end{pmatrix}.$$

(om deformationerna är *små*).

Härledning
Kraftsamband

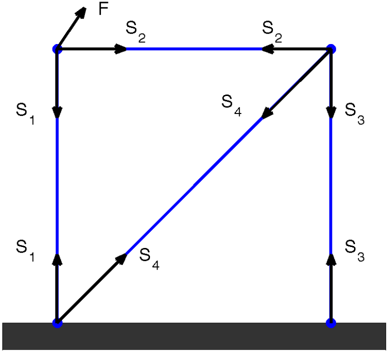


- Inför de inre krafterna S_1, S_2, S_3 och S_4 i stängerna.
- Motverkar yttre drag- och tryckkrafter:
  - $S_j < 0$ — stången ihoptryckt
  - $S_j > 0$ — stången utdragen

Kraftjämvikt: $\sum S_j + F = 0$

För de övre två noderna:

$$
\begin{aligned}
F_x + S_2 &= 0,\\
F_y - S_1 &= 0,\\
S_2 + \frac{1}{\sqrt{2}} S_4 &= 0,\\
S_3 + \frac{1}{\sqrt{2}} S_4 &= 0.
\end{aligned}
$$

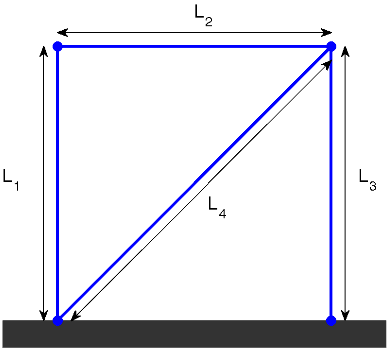


Hookes lag ger relationen mellan S_j och förlängning av stång ΔL_j:

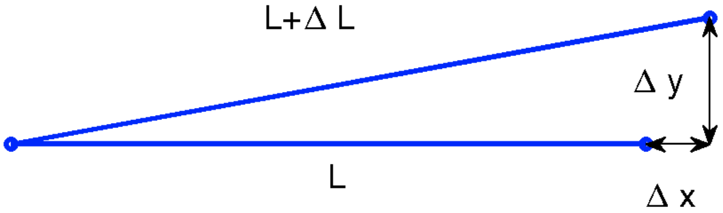
$$\frac{\Delta L_j}{L_j} = kS_j, \qquad k = \frac{1}{\text{elasticitetsmodul} \times \text{tvärsnittsarea}}$$

Vi antar att $k = L_1 = L_2 = L_3 = 1$ och $L_4 = \sqrt{2} \quad \Rightarrow$

$$\begin{aligned} \Delta L_2 &= -F_x, \\ \Delta L_1 &= F_y, \\ \Delta L_2 + \frac{1}{2}\Delta L_4 &= 0, \\ \Delta L_3 + \frac{1}{2}\Delta L_4 &= 0. \end{aligned}$$



Geometri ger relationen mellan ΔL och $(\Delta x, \Delta y)$.



Om ΔL, Δx, $\Delta y \ll L$, har vi

$$(L + \Delta L)^2 = (L + \Delta x)^2 + (\Delta y)^2 \quad \Rightarrow$$

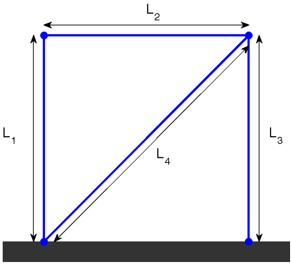
$$\Delta L = \Delta x + O\left(\frac{\Delta L^2 + \Delta x^2 + \Delta y^2}{L}\right) \approx \Delta x.$$

Eftersom $\Delta L \approx \Delta x$ får vi

$$\Delta L_1 \approx \Delta y_v, \qquad \Delta L_2 \approx \Delta x_h - \Delta x_v,$$

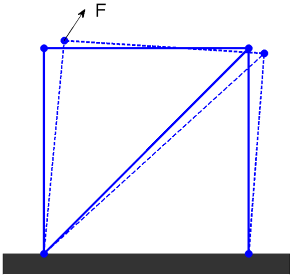
$$\Delta L_3 \approx \Delta y_h, \qquad \Delta L_4 \approx \frac{1}{\sqrt{2}}(\Delta x_h + \Delta y_h).$$

$$\begin{aligned}
\Delta x_h - \Delta x_v &= -F_x, \\
\Delta y_v &= F_y, \\
\Delta x_h - \Delta x_v + \frac{1}{2\sqrt{2}}(\Delta x_h + \Delta y_h) &= 0, \\
\Delta y_h + \frac{1}{2\sqrt{2}}(\Delta x_h + \Delta y_h) &= 0.
\end{aligned}$$



Linjärt system

Systemet



$$\Delta x_h - \Delta x_v = -F_x,$$
$$\Delta y_v = F_y,$$
$$\Delta x_h - \Delta x_v + \frac{1}{2\sqrt{2}}(\Delta x_h + \Delta y_h) = 0,$$
$$\Delta y_h + \frac{1}{2\sqrt{2}}(\Delta x_h + \Delta y_h) = 0.$$

kan skrivas

$$\begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 1+\frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} \\ 0 & 0 & \frac{1}{2\sqrt{2}} & 1+\frac{1}{2\sqrt{2}} \end{pmatrix} \begin{pmatrix} \Delta x_v \\ \Delta y_v \\ \Delta x_h \\ \Delta y_h \end{pmatrix} = \begin{pmatrix} F_x \\ F_y \\ 0 \\ 0 \end{pmatrix}.$$

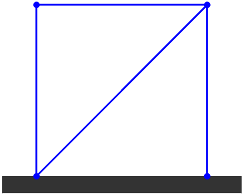
Ex (bild): $\boldsymbol{F} = (0.10,\, 0.15)$ ger

$$(\Delta x_v, \Delta y_v) \approx (0.48,\, 0.15), \qquad (\Delta x_h, \Delta y_h) \approx (0.38,\, -0.10).$$

Slutsatser

- I slutändan har vi

$$A = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 1+\frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} \\ 0 & 0 & \frac{1}{2\sqrt{2}} & 1+\frac{1}{2\sqrt{2}} \end{pmatrix}$$



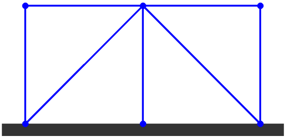
- Systemets storlek:

2 fria noder $\Rightarrow$ 4 obekanta $\Rightarrow$ $A \in \mathbb{R}^{4\times 4}$.

Slutsatser

- Precis samma princip för fackverk med fler noder. Ger $Ax = b$

$$A = \begin{pmatrix} 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 2+\frac{1}{\sqrt{2}} & 0 & -1 & 0 \\ 0 & 0 & 0 & 1+\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$



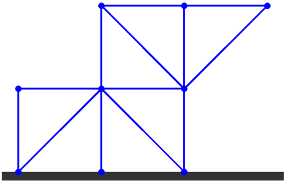
- Systemets storlek:

$$3 \text{ fria noder} \quad \Rightarrow \quad 6 \text{ obekanta} \quad \Rightarrow \quad A \in \mathbb{R}^{6\times 6}.$$

Slutsatser

- Precis samma princip för fackverk med fler noder. Ger $A\boldsymbol{x} = \boldsymbol{b}$

$$A = \begin{pmatrix}
1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-1 & 0 & 2.7071 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 2.7071 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & -1 & 0 & 1.7071 & 0 & -0.3536 & 0.3536 & 0 & 0 & -0.3536 & -0.3536 \\
0 & 0 & 0 & 0 & 0 & 2.7071 & 0.3536 & -0.3536 & 0 & -1 & -0.3536 & -0.3536 \\
0 & 0 & 0 & 0 & -0.3536 & 0.3536 & 1.3536 & -0.3536 & -1 & 0 & 0 & 0 \\
0 & 0 & 0 & -1 & 0.3536 & -0.3536 & -0.3536 & 1.3536 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 2.0000 & 0 & -1 & 0 \\
0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & -0.3536 & -0.3536 & 0 & 0 & -1 & 0 & 1.3536 & 0.3536 \\
0 & 0 & 0 & 0 & -0.3536 & -0.3536 & 0 & 0 & 0 & 0 & 0.3536 & 0.3536
\end{pmatrix}$$

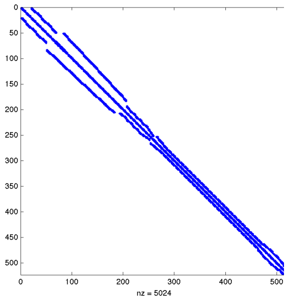


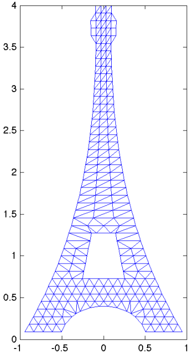
- Systemets storlek:

$$6 \text{ fria noder} \quad \Rightarrow \quad 12 \text{ obekanta} \quad \Rightarrow \quad A \in \mathbb{R}^{12\times 12}.$$

Slutsatser

- Precis samma princip för fackverk med fler noder. Ger $\boldsymbol{Ax} = \boldsymbol{b}$



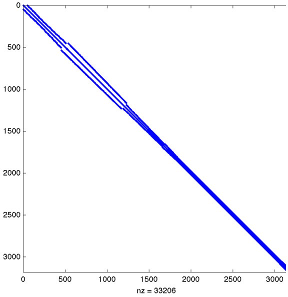


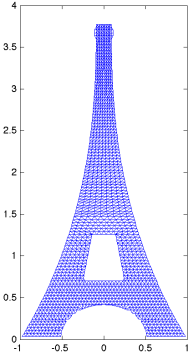
- Lab 1: Minsta modellen `eiffel1.mat`

$$261 \text{ fria noder} \quad \Rightarrow \quad 522 \text{ obekanta} \quad \Rightarrow \quad A \in \mathbb{R}^{522\times 522}.$$

Slutsatser

- Precis samma princip för fackverk med fler noder. Ger $\boldsymbol{Ax} = \boldsymbol{b}$





- Lab 1: Största modellen

$$4330 \text{ fria noder} \quad \Rightarrow \quad 8660 \text{ obekanta} \quad \Rightarrow \quad A \in \mathbb{R}^{8660 \times 8660}.$$

- Allmänt get n fria noder $2n$ obekanta och en styvhetsmatris $A \in \mathbb{R}^{2n \times 2n}$ som är gles.